



UNIVERSITY OF ASIA PACIFIC

DEPARTMENT OF CSE

COURSE CODE: CSE 402

COURSE TITLE: Operating System Lab

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Section: C-1

Semester: 4-1

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Lab Report: FCFS CPU Scheduling

Problem Statement

Calculate Completion Time (CT), Turnaround Time (TAT), Waiting Time (WT), Average Turnaround Time and Average Waiting Time using the FCFS scheduling algorithm. Also represent the process execution sequence.

Objective

The objective of this lab is to understand the First Come First Serve (FCFS) CPU scheduling algorithm and calculate CT, TAT and WT for the given processes. The average turnaround time, average waiting time and the execution sequence are also determined.

Given Data

Process ID	Arrival Time (AT)	Burst Time (BT)
P1	3	5
P2	2	4
P3	4	3
P4	1	2
P5	5	3

Working Principle

In FCFS scheduling, the process that arrives first gets the CPU first. The processes are arranged according to their arrival time. Completion Time is found from the execution order. Then TAT is calculated by $CT - AT$ and WT is calculated by $TAT - BT$.

Source Code

```
processes = [  
    ["P1", 3, 5],  
    ["P2", 2, 4],  
    ["P3", 4, 3],  
    ["P4", 1, 2],  
    ["P5", 5, 3]  
]  
  
order = sorted(processes, key=lambda x: x[1])  
  
result = {}  
current_time = 0  
execution = []  
  
for p in order:  
    pid = p[0]  
    at = p[1]  
    bt = p[2]  
  
    if current_time < at:  
        current_time = at
```

```

    ct = current_time + bt
    tat = ct - at
    wt = tat - bt

    current_time = ct

    result[pid] = [ct, tat, wt]
    execution.append(pid)

print("P_ID   AT   BT   CT   TAT   WT")

for p in processes:
    pid = p[0]
    at = p[1]
    bt = p[2]

    ct = result[pid][0]
    tat = result[pid][1]
    wt = result[pid][2]

    print(f"{pid:<7}{at:<5}{bt:<5}{ct:<5}{tat:<6}{wt}")

total_tat = 0
total_wt = 0

for p in processes:
    total_tat += result[p[0]][1]
    total_wt += result[p[0]][2]

avg_tat = total_tat / len(processes)
avg_wt = total_wt / len(processes)

print()
print("AVG TAT =", avg_tat)
print("AVG WT  =", avg_wt)

print()
print("Execution Sequence:")
print(" -> ".join(execution))

```

Output

```
Run FCFS x
C:\Users\HP\PycharmProjects\lab1\.venv\Scripts\python.exe C:\Users\HP\PycharmProjects\lab1\FCFS.py

P_ID  AT  BT  CT  TAT  WT
P1    3   5  12   9   4
P2    2   4   7   5   1
P3    4   3  15  11   8
P4    1   2   3   2   0
P5    5   3  18  13  10

AVG TAT = 8.0
AVG WT  = 4.6

Execution Sequence:
P4 -> P2 -> P1 -> P3 -> P5

Process finished with exit code 0
```

Discussion

The FCFS algorithm executes the processes according to their arrival time. For the given data, the execution order is P4, P2, P1, P3 and P5. The completion time, turnaround time and waiting time were calculated for each process. The average turnaround time is 8.0 and the average waiting time is 4.6. This experiment helped to understand how FCFS scheduling works and how process arrival time affects the execution order and waiting time.

Conclusion

The FCFS scheduling algorithm was implemented successfully in Python. The required CT, TAT, WT, average TAT, average WT and execution sequence were obtained correctly. From this experiment, the basic working procedure of FCFS scheduling became clear. It also showed that processes are executed in the order of their arrival time and each process waits until the previous process finishes.